

二. (20 points) Three processes P1、P2 and P3 using a buffer having $N(N>0)$ units mutually exclusively. For each time, P1 uses `produce()` to generate a non-negative integer and then uses `put()` to send the integer into an empty unit in the buffer; P2 uses `getodd()` to pick one odd number from the buffer and then uses `countodd()` to count the number of odd numbers; P3 uses `geteven()` to pick one even number from the buffer and then uses `counteven()` to count the number of even numbers. Please implement the synchronization and mutual exclusion activities of these three processes using the semaphore mechanism and describe the meaning of the semaphores you defined. It requires a pseudo-code description.

三.(15 points) Consider the following snapshot of a system, answer the following questions according to the Banker's algorithm.

Allocation					MAX				
	A	B	C	D		A	B	C	D
P0	0	0	1	2	P0	0	0	3	2
P1	2	0	0	0	P1	2	7	5	0
P2	0	0	3	4	P2	6	6	5	6
P3	2	3	5	4	P3	4	3	5	6
P4	0	3	3	2	P4	0	6	5	2

Available			
A	B	C	D
2	1	2	0

- (1) Calculate matrix NEED. Is the system in safe state? Why?
- (2) If a request from process P2 arrives for resources (0,1,2,0), can the request be granted immediately? If yes, then show the sequence of successful process executions in tabular form; if no, then you do not need to show any further details.

四. (15 points) In a paging management system, assume that the contents of the page table for a process are shown in the following table:

Page number	Page frame number	Valid bit
0	202H	1
1	-	0
2	254H	1

The page size is 4 KB, a memory access time is 120 ns, a fast table (TLB) access time is 20 ns, the average time to process a missing page is 10^7 ns (including the time to update the TLB and page table), the process has a fixed resident set size of 2, and uses the least recent use replacement algorithm (LRU) and the local elimination policy. Assumptions: 1. The TLB is initially empty; 2. The TLB is accessed first during address translation, and if the TLB does not hit, then the page table is accessed (ignoring the TLB update time after the page table is accessed); 3. A valid bit of 0 indicates that the page is not in a memory page, and a missing-page interrupt is generated, and the missing-page interrupt is processed by returning to the instruction that generated the missing-page interrupt to be re-executed. There is a virtual address access sequence 2362H, 1566H and 25A5H, please answer the following questions:

- (1) How long does it take to access each of the above three virtual addresses in turn?
- (2) What is the physical address of virtual address 1565H, please give your reason.

五.(20 points) A file system has a disk block size of 4 KB, and the directory entries consist of filenames and index node numbers, each index node occupies 256 bytes, including 10 direct address entries, and one each of first-, second-, and third-level indirect address entries, each of which occupies 4 bytes. The structure of the subdirectory *stu* in the file system is shown in Figure 1, *stu* contains the subdirectory *course* and the file *doc*, and the *course* subdirectory contains the file *course1* *course2*. The file name, index node number, and block number of the occupied disk block of each file are shown in Figure 2.

Answer the following questions.

- 1) What is the content of each directory entry in the directory file *stu*?
- 2) What is the value of the block number *x* of the disk block occupied by the file *doc*?
- 3) If the contents of the directory file *course* are already in memory, how many disk blocks are required to open the file *course1* and read it into memory? Explain.
- 4) If the size of the file *course2* grows to 6MB, how many disk blocks are needed to store *course2*? If the size of the file *course2* grows to 6 MB, what level of indirect address entries in the index node of the file will be used to retrieve *course2*? Give reasons.

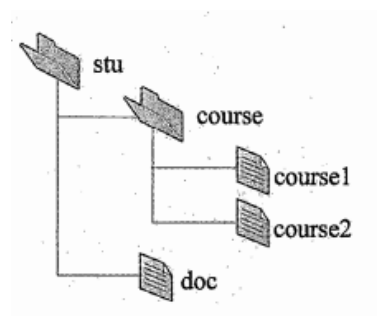


Figure 1

File name	Index node number	Disk block number
stu	1	10
course	2	20
course1	10	30
course2	100	40
doc	10	x

Figure2

六.(15 points) A computer system uses a CSCAN (Cyclic Scanning) disk scheduling policy to record the idle status of 16,384 disk blocks using 2 KB of memory space.

(1) Explain how the management of disk block idle states is performed under the above conditions.

(2) A single-sided disk rotates at 6,000 revolutions per minute, each track has 100 sectors, and the average travel time between adjacent tracks is 1ms. If at a certain point in time the magnetic head is located at track 100 and is moving in the direction of the increasing track number, and the queue of track number requests is 50, 90, 30, and 120, and one randomly distributed sector needs to be read from each track in the queue, then the read time for each randomly distributed sector in the queue will be 1ms, and the read time for each randomly distributed sector will be 1ms. The request queue is 50, 90, 30, 120,130, and 1 randomly distributed sector needs to be read for each sector in the request queue, how much time will it take to read these 4 sectors in total? Requirements to give the calculation process.